ALGORITHM: MUSICAL PLAYLIST MANAGER

Data Structure: Doubly Linked List

Each Song node stores title, artist, album, and pointers to the previous and next songs.

1. Initialize Playlist

• Set head = NULL, tail = NULL, and current = NULL.

• Display the menu and read the user's choice.

2. Add Song

• Read the song title, artist, and album.

• Create a new Song node.

• If the playlist is empty, set head, tail, and current to the new node.

• Otherwise, link the new node after tail and update tail.

3. Display Playlist

• If head is NULL, display that the playlist is empty.

• Otherwise, traverse from head using next and print each song.

4. Play Current Song

• If current is NULL, display that the playlist is empty.

• Otherwise, display the current song's title, artist, and album.

5. Play Next Song

• If current is NULL, display that the playlist is empty.

• If current->next is NULL, display that the last song is already selected.

• Otherwise, set current = current->next and display the song.

6. Play Previous Song

• If current is NULL, display that the playlist is empty.

• If current->prev is NULL, display that the first song is already selected.

• Otherwise, set current = current->prev and display the song.

7. Search Song

• Read the title to search for.

• Traverse from head and compare each node's title.

• If found, display the song details; otherwise, display Song not found.

8. Delete Song

• Read the title and locate the matching node.

• Reconnect the previous and next nodes; update head or tail if needed.

• If current is the deleted node, move current to the next node if possible, otherwise to the previous node.

• Delete the node and display a message.

9. Reverse Playlist

• If the playlist is empty, display a message.

• For each node, swap its prev and next pointers.

• Swap head and tail.

10. Clear Playlist

• Traverse the list and delete each node, saving next before deletion.

• Set head, tail, and current to NULL.

11. Exit

• Repeat the menu until the user chooses 0.

• The destructor deletes any remaining nodes.

• End the program.